Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/14/2006
PI Name:	Dionysiou, Dionysios Ph.D.		
Project Title:	High Performance TiO2 Photocatalytic Coatings and Membranes for the Purification, Disinfection and Recycle of Water and Air in Space Applications		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Dionysios.D.Dionysiou@UC.Edu	**Fax:**	FY 513-556-2599
PI Organization Type:	UNIVERSITY	**Phone:**	513-556-0724
Organization Name:	University of Cincinnati		
PI Address 1:	Department of Civil & Environmental Engineering		
PI Address 2:	Mail Stop 71		
PI Web Page:	http://www.eng.uc.edu/		
City:	Cincinnati	**State:**	OH
Zip Code:	45221-0071	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	01/14/2003	**End Date:**	01/14/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: received NCE through 1/14/2007, per PI 5/2006		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
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Task Description:	This study deals with the application of novel methods for the fabrication of robust nanostructured TiO2 photocatalytic films for the development of efficient water and air purification systems for advanced life support in space missions and terrestrial applications. The catalytic films immobilized on a variety of ceramic, metallic, and composite substrates will be used to fabricate reactors and membrane systems that will pose unique pore orchestration for the effective capture of incident radiation, enhanced photocatalytic activity, excellent adhesive properties, good mechanical stability, controlled hydrophilicity, and anti-biological fouling capability. These immobilized films will be the crucial component for the fabrication of integrated photocatalytic-disinfection-antifoulant reactors and membranes for the room-temperature production of clean potable and hygienic water in space missions as well as for the purification of water in terrestrial applications. Alternatively, these methods will be used for the fabrication of immobilized TiO2 films aiming at the preparation of surfaces possessing self-cleaning, disinfecting, and anti-biofouling properties for air purification. These purification systems can be used as stand-alone technologies or as supplementary and complementary to existing treatment technologies. In addition, they can serve as back-up processes or as a means of emergency-respond technologies to chemical and biological terrorism. We apply a novel fabrication procedure for the preparation of high performance TiO2 films immobilized on non-porous stainless steel substrates. The procedure concerns a novel combination of the alkoxide sol-gel and the colloidal suspension (Degussa P-25) preparation routes for the synthesis of the catalyst. We optimize the P-25 concentration and calcination temperature for enhanced photocatalytic activity and good mechanical stability. The purpose of this research is to better understand the relationship of surface microstructure and film composition and their effect on the photocatalytic activity of the films. After the optimization of the modified sol-gel procedure, we evaluate the photocatalytic activity of these films for the degradation of target contaminants. Our goal is to optimize the conditions for the destruction and mineralization of these compounds with prepared TiO2 photocatalyst and to identify any inhibition factors. Another goal is the design of a new reactor with higher surface area per unit liquid volume treated, increased photocatalytic activity, and better optimization for efficient utilization of UV radiation. In addition, we apply novel nanofabrication and self-assembling procedures to prepare highly porous nanostructured photocatalytic TiO2 particles, films immobilized on nonporous glass substrates, and membranes incorporated onto porous alumina membrane supports. The objective for this task is to use self-assembling modified sol-gel methods for the fabrication of TiO2 thin films of controlled intraparticle (framework) and interparticle (textural) porosity to be used on top of non-porous or porous supports. In the latter case (i.e., porous supports), the method is used for the fabrication of nanocomposite TiO2-ceramic membranes with tailor-designed pore size (mesoporous, 1-50 nm), improved photocatalytic properties, and enhanced mechanical stability. This research is considered as an important task within our scope to develop high performance photocatalytic coatings for water purification. The completion of the tasks and objectives is crucial for the fabrication of highly efficient robust integrated advanced oxidation reactors for the purification, disinfection, and recycle of water in space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The materials prepared in this project and the associated technologies are of interest to the public as means to provide clean water and air and thus to protect human health. In addition, special educational activities focus on public education on issues relevant to nanoscience and nanotechnology. In addition to the development of novel photocatalytic coatings and membranes for water and air treatment, purification and disinfection for Advanced Life Support in space missions, this study can lead to the development of robust photocatalytic water and air purification systems for terrestrial applications. The study can also enhance fundamental knowledge for the fabrication of novel ceramic materials, films, and membranes with anti-biofouling properties. The method for the preparation of films with super-hydrophilic properties can be applied to other support materials with desirable properties for water spreading, light transmittance, and self-cleaning properties.
	The proposed work deals with the application of novel aspects on TiO2 photocatalysis for the fabrication of robust nanostructured TiO2 photocatalytic films and membranes. These films and membranes will be an inherent component of efficient water and air purification systems suitable to provide advanced life support in long-duration space missions of NASA as well as in long-term terrestrial applications. These purification systems can be used as stand-alone technologies or as supplementary and complementary to existing treatment technologies. In addition, they can serve as back-up processes or as a means of emergency-respond technologies to chemical and biological terrorism. The project concerns various innovative approaches for the development of highly active and mechanically stable immobilized TiO2 coatings. In the third year, we continued to conduct the research on the synthesis of TiO2 photocatalytic films and membranes and their environmental applications. First, we have focused on the synthesis of high performance TiO2 photocatalytic coating with robust crystalline structure and good structural integrity and its application in decomposing human metabolites. The best conditions for the synthesis of TiO2 films immobilized on stainless steel using the combined alkoxide sol-gel and colloidal suspension preparation routes were investigated. Our previous work focused on the optimization of certain preparation parameters, including P25 loading and calcination temperatures. In the third year, the preparation of crack-free photocatalytic coatings was investigated The preparation method includes employing lower loadings of P25 power in the sol and improving the structure of TiO2 matrix by using templates as pore formation agents. We have made a comparative study on the physicochemical properties and photocatalytic behavior of porous TiO2-P25 composite films and porous TiO2 films coated on stainless steel. The results showed that incorporating low loading P25 into the sol can effectively decrease or avoid crack formation in the P25 powder modified TiO2 film. The photocatalytic activity of TiO2 matrix can be remarkably improved by using polyethylene glycol 2000 as a template. Then, the evaluation of the TiO2 photocatalytic films prepared with a combined alkoxide-colloidal suspension sol-gel method was performed with creatinine. Creatine was found to be the major intermediate during the photocatalysis of creatinine at acidic conditions. Since the hydrolysis of peptidic bonds occurs in alkali pHs, hydroxyl radicals are utilized for breaking this bond, forming the linear aminoacid creatine at pH= 3.0. The formation of NOx- was also monitored. In addition, we investigated new possible substrates for the photocatalytic coatings. Porous stainless steel materials (sieves) with different mess sizes are almost ten times lighter than the stainless steel sheets and contain eight times higher catalyst loading. The sieves gave better purification and mineralization efficiencies than the stainless steel substrates. We are currently utilizing other methods to coat the sieves for the preparation of crack-free films with good mechanical stability. Second, our research studies have been carried out to immobilize TiO2 catalyst onto porous substrates for fabricating TiO2 membranes for use in a variety of applications. In our previous study during the first and second years, we described the application of novel chemistry methods for the fabrication of robust nanostructured TiO2 photocatalysts.

Task Progress:	We also demonstrated the multiple functions of the TiO2 membranes, such as (i) decomposition of recalcitrant organic pollutants, (ii) inactivation and killing of pathogenic microorganisms, and (iii) physical separation of contaminants. However, repetition of the dip-coating procedure using a single sol in the study resulted in the thick symmetric TiO2 coating layer, which increased hydraulic resistance through the membrane. The ideal membrane structure would be composed of a thin asymmetric multilayer with increasing pore size and porosity from the top to the bottom of TiO2 skin layer. Consequently, during the third year, our concern was to fabricate nanocrystalline TiO2 photocatalytic membranes with a robust hierarchical mesoporous multilayer and improved performance, where the resulting asymmetric TiO2 membrane exhibited a hierarchical change in pore diameter from 2 - 6 nm, 3 - 8 nm, to 5 - 11 nm from the top to the bottom layer. Moreover, the corresponding porosity was incremented from 46.2 %, 56.7 % to 69.3 %. Compared to a repeated-coating process using a single sol, which was developed during the previous research, the hierarchical multilayer process improved water permeability significantly, without sacrificing organic retention and photocatalytic activity of the TiO2 membranes. In this method, the surfactant concentration largely affected the structural properties of the TiO2 material. Controlling materials at the nano-level made it possible to develop new types of catalytic membrane products with tailor-designed properties exhibiting a hierarchical change in the diameter of the mesopore from the top to the bottom of the TiO2 skin layer. In general, the results obtained in this work so far are promising and are expected to be of high interest to NASA, the scientific community, and many relevant industries for the development of sustainable photocatalytic reactors as well as for the development of coatings with task-specific applications. The good structural properties and enhanced catalytic activities of the prepared TiO2 materials are important requirements for the fabrication of sustainable photocatalytic reactors for water and air purification. In addition to physical research activities, the project has provided valuable opportunities to graduate students. The students had an opportunity to learn novel procedures for the fabrication of nanostructured TiO2 films and membranes and exchange valuable experience and knowledge in the area of materials synthesis and their environmental applications. Students in the Environmental Science and Engineering program at the University of Cincinnati also had the opportunity to learn current issues in the field of nanotechnology and became familiar with the synthesis, characterization, and environmental applications and implications of nanomaterials. Dr. Dionysiou and his group also promoted awareness on the health threat of environmental pollution in water and wastewater as well as the importance of environmental preservation to people, including public and environmental affairs around the Cincinnati regional area. In order to disseminate the importance of nanotechnology in the area of environmental remediation to colleagues, public, and communities, Dr. Dionysiou and his students have presented their research findings on the degradation of hazardous organic contaminants using TiO2 photocatalytic films and membranes at several regional, national, and international conferences and meetings and they were acknowledged with several awards, fellowships, and scholarships. Finally, Dr. Dionysiou's collaboration with other national and international universities, expanded the opportunities for advanced and international level research.
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